

## SALEM 10 W, MO, USA

WMO: 725600

Lat: 37.6344N

Lon: 91.7226W

Elev: 365

StdP: 97.01

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23909

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.2      | -12.1 | -20.0                         | 0.7 | -14.3 | -17.4 | 0.9 | -10.3 | 6.5                   | 9.5  | 5.8 | 7.5  | 1.4                   | N/A  | 0.408 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.6                   | 34.4            | 23.0 | 32.6 | 23.2 | 31.2 | 23.0 | 25.2                | 30.9 | 24.6 | 30.2 | 23.9 | 29.3 | 2.2                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.6                            | 19.3 | 28.4 | 22.9 | 18.4 | 27.8 | 22.2 | 17.7 | 26.9 | 79.2          | 30.9 | 76.4 | 30.1 | 73.5 | 29.4 | 27.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 6.0           | 5.3  | 4.6 | DB                         | -21.3 | 36.3               | 5.0 | 2.4                                                | -24.9 | 38.0       | -27.8 | 39.4       | -30.6 | 40.7       | -34.2 | 42.4 | (4) |
| (5)               |      |     | WB                         | -22.1 | 26.2               | 5.2 | 1.0                                                | -25.9 | 26.9       | -28.9 | 27.5       | -31.9 | 28.0       | -35.7 | 28.7 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.6   | -0.4  | 1.7   | 7.5   | 13.2  | 17.5  | 22.2  | 23.9  | 23.2  | 19.5  | 13.3  | 6.8   | 1.9  |
|                                                                                  | DBStd       | 9.90   | 6.69  | 6.69  | 6.31  | 5.11  | 4.59  | 3.07  | 3.02  | 3.18  | 3.97  | 5.27  | 5.55  | 5.59 |
|                                                                                  | HDD10.0     | 1133   | 328   | 243   | 124   | 26    | 4     | 0     | 0     | 0     | 0     | 26    | 126   | 257  |
|                                                                                  | HDD18.3     | 2696   | 582   | 464   | 339   | 166   | 71    | 7     | 1     | 3     | 34    | 174   | 346   | 509  |
|                                                                                  | CDD10.0     | 2076   | 5     | 12    | 45    | 121   | 237   | 367   | 432   | 409   | 285   | 127   | 30    | 6    |
|                                                                                  | CDD18.3     | 599    | 0     | 0     | 3     | 12    | 47    | 123   | 176   | 153   | 69    | 17    | 1     | 0    |
|                                                                                  | CDH23.3     | 6073   | 0     | 2     | 33    | 135   | 458   | 1223  | 1818  | 1538  | 706   | 152   | 7     | 0    |
|                                                                                  | CDH26.7     | 2316   | 0     | 0     | 4     | 23    | 109   | 433   | 802   | 669   | 248   | 27    | 1     | 0    |
| Wind                                                                             | WSAvg       | 1.9    | 2.1   | 2.3   | 2.5   | 2.7   | 2.1   | 1.7   | 1.3   | 1.3   | 1.2   | 1.6   | 1.9   | 2.0  |
| Precipitation                                                                    | PrecAvg     | 1174   | 66    | 65    | 99    | 117   | 142   | 104   | 102   | 98    | 112   | 83    | 101   | 77   |
|                                                                                  | PrecMax     | 1533   | 145   | 159   | 273   | 317   | 320   | 211   | 283   | 220   | 322   | 268   | 244   | 252  |
|                                                                                  | PrecMin     | 689    | 14    | 11    | 28    | 19    | 22    | 35    | 32    | 42    | 11    | 7     | 18    | 17   |
|                                                                                  | PrecStd     | 238    | 31    | 38    | 51    | 61    | 65    | 49    | 54    | 41    | 73    | 52    | 58    | 51   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 19.3  | 22.7  | 26.8  | 29.3  | 31.6  | 34.8  | 37.0  | 36.9  | 33.2  | 29.6  | 24.1  | 19.4 |
|                                                                                  |             | MCWB   | 12.5  | 15.0  | 16.3  | 17.8  | 19.6  | 22.8  | 22.0  | 23.3  | 22.6  | 20.6  | 15.5  | 14.1 |
|                                                                                  | 2%          | DB     | 16.5  | 19.2  | 23.7  | 26.7  | 29.3  | 31.8  | 34.6  | 34.4  | 31.4  | 26.9  | 21.2  | 16.6 |
|                                                                                  |             | MCWB   | 10.4  | 12.5  | 15.1  | 16.7  | 20.3  | 23.5  | 23.0  | 23.4  | 22.6  | 18.5  | 13.6  | 12.9 |
|                                                                                  | 5%          | DB     | 13.1  | 16.4  | 20.5  | 24.8  | 27.8  | 30.6  | 32.5  | 32.5  | 29.8  | 25.1  | 18.7  | 13.9 |
|                                                                                  |             | MCWB   | 8.4   | 11.4  | 12.9  | 15.9  | 19.9  | 23.2  | 23.7  | 22.9  | 22.0  | 17.4  | 13.2  | 9.5  |
|                                                                                  | 10%         | DB     | 10.1  | 13.2  | 18.1  | 22.6  | 26.1  | 29.4  | 31.0  | 30.5  | 27.7  | 22.5  | 16.6  | 11.4 |
|                                                                                  |             | MCWB   | 6.5   | 8.9   | 11.7  | 15.1  | 19.2  | 22.6  | 23.4  | 22.7  | 20.7  | 16.2  | 11.4  | 7.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.1  | 16.2  | 17.8  | 20.0  | 22.8  | 25.6  | 26.1  | 26.0  | 24.1  | 21.8  | 18.0  | 16.0 |
|                                                                                  |             | MCDB   | 18.0  | 21.0  | 24.5  | 26.5  | 28.7  | 31.0  | 31.5  | 32.3  | 30.1  | 27.4  | 21.7  | 18.5 |
|                                                                                  | 2%          | WB     | 11.8  | 14.7  | 16.2  | 18.7  | 21.9  | 24.7  | 25.3  | 25.1  | 23.4  | 20.7  | 15.8  | 13.6 |
|                                                                                  |             | MCDB   | 14.4  | 18.4  | 22.2  | 24.2  | 27.2  | 30.3  | 30.9  | 31.0  | 29.3  | 24.6  | 19.1  | 15.9 |
|                                                                                  | 5%          | WB     | 9.1   | 12.3  | 14.5  | 17.5  | 21.1  | 24.0  | 24.7  | 24.2  | 22.7  | 19.1  | 13.9  | 10.8 |
|                                                                                  |             | MCDB   | 12.0  | 14.8  | 19.1  | 22.3  | 26.0  | 29.4  | 30.3  | 29.7  | 28.3  | 22.4  | 17.5  | 13.1 |
|                                                                                  | 10%         | WB     | 6.5   | 8.6   | 12.8  | 16.4  | 20.2  | 23.2  | 24.0  | 23.3  | 21.8  | 17.7  | 11.8  | 8.5  |
|                                                                                  |             | MCDB   | 10.1  | 12.9  | 16.4  | 21.4  | 24.9  | 28.0  | 29.4  | 28.4  | 26.8  | 21.6  | 15.8  | 10.5 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.0  | 11.9  | 13.0  | 13.2  | 12.3  | 12.0  | 12.6  | 12.8  | 13.4  | 13.7  | 13.0  | 11.3 |
|                                                                                  |             | MCDBR  | 16.2  | 16.1  | 16.5  | 15.4  | 14.5  | 13.5  | 14.5  | 15.5  | 14.9  | 15.1  | 15.5  | 15.3 |
|                                                                                  |             | MCWBR  | 10.8  | 11.2  | 9.0   | 7.7   | 6.8   | 6.6   | 5.4   | 5.8   | 6.9   | 7.7   | 9.6   | 10.1 |
|                                                                                  | 5% WB       | MCDBR  | 13.0  | 14.4  | 13.7  | 12.8  | 12.0  | 11.8  | 12.1  | 12.3  | 13.3  | 12.1  | 12.6  | 12.1 |
|                                                                                  |             | MCWBR  | 10.3  | 11.0  | 8.5   | 7.7   | 6.4   | 6.3   | 5.6   | 5.6   | 6.4   | 6.9   | 9.3   | 9.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.299  | 0.307 | 0.334 | 0.382 | 0.417 | 0.425 | 0.454 | 0.448 | 0.385 | 0.343 | 0.311 | 0.296 |      |
|                                                                                  | taud        | 2.493  | 2.464 | 2.435 | 2.307 | 2.262 | 2.265 | 2.215 | 2.226 | 2.382 | 2.491 | 2.529 | 2.516 |      |
|                                                                                  | Ebn at Noon | 892    | 929   | 930   | 898   | 867   | 857   | 829   | 825   | 864   | 874   | 871   | 871   |      |
|                                                                                  | Edh at Noon | 80     | 93    | 106   | 127   | 136   | 135   | 141   | 137   | 110   | 90    | 77    | 73    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.27   | 2.90  | 3.89  | 5.04  | 5.64  | 6.45  | 6.26  | 5.60  | 4.82  | 3.54  | 2.46  | 1.91  |      |
|                                                                                  | RadStd      | 0.23   | 0.34  | 0.35  | 0.30  | 0.50  | 0.47  | 0.42  | 0.30  | 0.31  | 0.39  | 0.26  | 0.22  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (22 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page